

Multiple Choice

1. **Answer:** B

Options: A) do the experiment many times; B) keep a record of all procedures; C) make a model of the set-up of the experiment; D) change some of the conditions of the experiment

Note: Keeping a record of all procedures is essential for ensuring that the experiment can be accurately replicated, allowing other scientists to verify results and understand the methodology used.

2. **Answer:** C

Options: A) unhappy.; B) confused.; C) confident.; D) generous.

Note: Leopard's opening chant reflects his self-assured demeanor and assertive vocalization, indicating a high level of confidence in his abilities and presence within his environment.

3. **Answer:** B

Options: A) They both are formed from the remains of dead organisms.; B) They both are acquired by mining the lithosphere.; C) They both release energy through combustion.; D) They both generate radioactive wastes.

Note: Both uranium and coal are natural resources that are extracted from the Earth's lithosphere through mining processes.

4. **Answer:** B

Options: A) an increase in animal populations; B) a decrease in oxygen production; C) an increase in precipitation; D) a decrease in pollutants

Note: Large areas of deforestation in the Amazon rainforest lead to a decrease in oxygen production because trees and plants, which absorb carbon dioxide and release oxygen through photosynthesis, are removed, reducing the overall capacity of the ecosystem to generate oxygen.

5. **Answer:** C

Options: A) There is less rainfall in the desert.; B) Temperatures vary more in the desert.; C) There is less water vapor in the desert air.; D) There are fewer plants in the desert.

Note: The correct answer is based on the fact that early morning fog forms when warm, moist air cools and condenses, and since desert air contains significantly less water vapor than coastal air, it rarely reaches the necessary humidity levels for fog formation.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The Moon's synchronous rotation, where it rotates on its axis in the same time it takes to orbit the Earth, is primarily due to gravitational interactions between the Earth and the Moon, not the gravitational pull of the Sun.

7. **Answer:** False

Options: A) True; B) False

Note: Fossil fuels are called nonrenewable resources because they take millions of years to form and cannot be replenished within a human timescale, not because of their location underground.

8. **Answer:** True

Options: A) True; B) False

Note: Biological magnification of pollutants is most pronounced in estuaries because these ecosystems are highly productive and support a diverse array of organisms, leading to higher concentrations of pollutants as they move up the food chain.

9. **Answer:** True

Options: A) True; B) False

Note: The Moon reflects sunlight, making it much dimmer than the Sun, which reduces the risk of eye damage when looked at directly.

10. **Answer:** True

Options: A) True; B) False

Note: A wildfire can rapidly destroy vegetation and disrupt habitats, leading to significant ecological changes within a single day due to the intense heat and flames that consume plant life and alter the environment.

Long Form Response

11. **Answer:** Data from similar studies can significantly strengthen the conclusions drawn by a lab group investigating plant growth rates near a classroom window. When researchers find consistent results across multiple studies, it increases the reliability of their findings, suggesting that the observed plant growth is not just a coincidence. For example, if other studies show that plants receive more sunlight near windows and grow faster as a result, this supports the lab group's conclusion about the impact of light on growth rates. By comparing their data with findings from other experiments, the lab group can better understand the factors influencing plant growth and make more confident claims about their observations. This collaborative approach to data reinforces the scientific method, highlighting the importance of gathering evidence from various sources.

Note: correct because it emphasizes that consistent findings from similar studies enhance the credibility of the lab group's conclusions by demonstrating that observed

plant growth patterns are likely due to the environmental factor of sunlight availability, rather than chance.

12. **Answer:** Decreasing the heat energy of a gas causes its molecules to move more slowly, which impacts its properties and behavior significantly. As the kinetic energy of the gas decreases, the temperature drops, leading to a reduction in pressure if the gas is allowed to expand. For example, when a balloon is placed in a refrigerator, the gas inside cools, causing the balloon to shrink as the gas pressure decreases. Additionally, if the gas is cooled enough, it may condense into a liquid, demonstrating how temperature changes can alter the state of matter. Overall, lowering the heat energy affects gas behavior by reducing molecular motion and altering its physical state.

Note: correct because decreasing the heat energy of a gas reduces the kinetic energy of its molecules, leading to lower temperature and pressure, which is exemplified by the behavior of a balloon in a refrigerator and the potential for gas condensation into liquid.